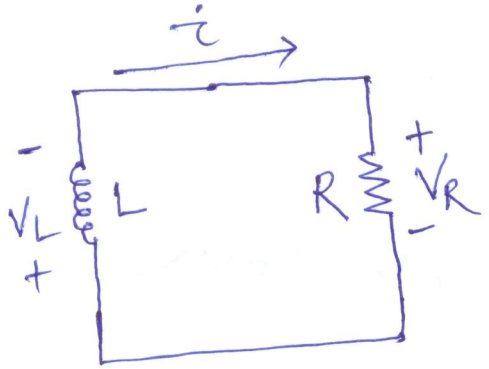
The Source-Free RL ckt

Consider a series connection of an inductor & Resistor.



by KVL, we have

$$V_L + V_R = 0$$

$$\Rightarrow L\frac{di(t)}{dt} + i(t)R = 0$$

$$\Rightarrow \int_{i(0)}^{i(t)} \frac{di(t)}{i(t)} = -\frac{R}{L}\int_0^t dt$$

$$\Rightarrow \left[\ln\{i(t)\}\right]_{i(0)}^{i(t)} = -\frac{R}{L}\left[t\right]_0^t$$

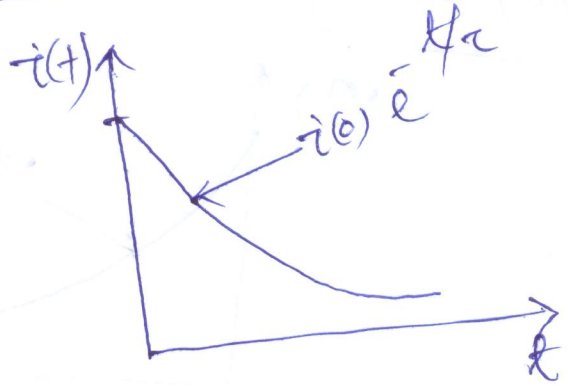
$$\Rightarrow \ln\left[\frac{i(t)}{i(0)}\right] = -\frac{R}{L}\times(t-0)$$

$$\Rightarrow e^{-\frac{R}{L}t}$$

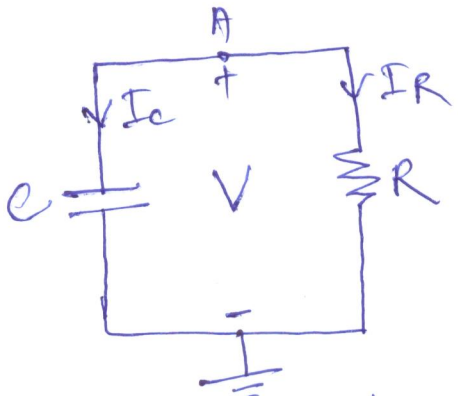
$$\Rightarrow i(t) = i(0)\,e^{-t/\tau}$$

$$\boxed{i(t) = i(0)\,e^{-t/\tau}}, \quad \tau = \frac{L}{R} = \text{time constant}$$

Response



Source-Free RC CKt



by KCL at node 'A', we have

$$I_c + I_R = 0$$

$$I_c = -I_R$$

$$\Rightarrow C\frac{dV}{dt} = -\frac{V}{R}$$

$$\Rightarrow \int \frac{dV}{V} = \int -\frac{1}{RC}\,dt$$

$$\Rightarrow \ln V = -\frac{1}{RC}t + K$$

$$V = e^{-\frac{1}{RC}t + K} = e^{-\frac{1}{RC}t} \cdot e^{K}$$

$$= e^{-\frac{1}{RC}t} \cdot K$$

$$V = K e^{-\frac{1}{RC}t}$$

for $t = 0$, $K = V(0)$

so $\boxed{V(t) = V(0)\, e^{-\frac{t}{\tau}}}$, $\tau = RC$ = time constant.

